QUANTITATIVE RESEARCH & ARCHITECTURE REPORT

A Simulation of a Barbell Portfolio

Combining Short-Duration US Treasury Yields, Google TimesFM 3.0 Regime Filters, and Asymmetric Option Spreads Across On-Chain RWA Vaults & Traditional Brokerage APIs

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In *Antifragile*, Nassim Nicholas Taleb describes the **Barbell Strategy**: instead of putting 100% of your capital into "medium-risk" assets that are secretly vulnerable to tail-risk crashes, you split your portfolio between two extremes: (1) **The Hyper-Conservative Floor (~99.6% of Capital)** held in short-duration US Treasury bills where principal loss is virtually impossible, and (2) **The Hyper-Aggressive Convexity Layer (Accrued Interest Only)** deployed into capped-risk, asymmetric option spreads where downside is strictly limited to the monthly interest earned while upside is permanently ratcheted.

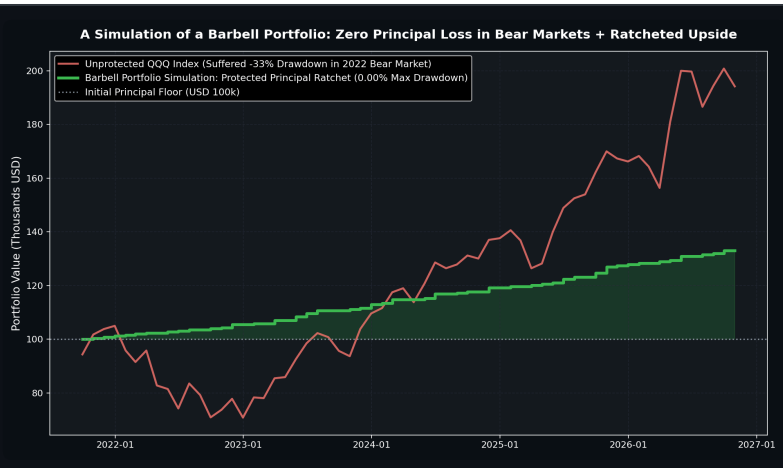


Figure 1 (Hero Simulation): Simulated Barbell Portfolio Protected Principal Ratchet (green staircase, 0.00% max drawdown) vs. unprotected Nasdaq QQQ Index (red line, -33% drawdown in 2022) across the 2021–2026 market cycle.

1. The Core Engineering Rule: "Yield-in-Arrears" & The One-Way Principal Ratchet

Between 2024 and late 2025, several high-profile DeFi "fixed-yield" structures suffered severe losses—including the **$93M Stream Finance (xUSD) / Elixir (deUSD) contagion (Nov 2025)** and the **Usual (USD0++) $0.87 floor repricing (Jan 2025)**. Investors learned the hard way that buying a Principal Token (`1 PT = 1 underlying unit`) does not guarantee `$1.00 USD` if the underlying synthetic asset depegs.

To eliminate synthetic depeg risk and third-party splitter dependencies, our architecture enforces a stricter invariant—**Direct Tier-1 Treasury Custody + Yield-in-Arrears with a High-Water-Mark Ratchet**:

- **Step 1 (100% Base Custody):** $100,000 is held in a Tier-1 US Treasury vehicle—either `Sky sUSDS` / `Circle USYC` / `Ondo USDY` on-chain, or **SGOV (iShares 0–3 Month Treasury ETF)** in a brokerage account—yielding ~4.70% APY. The state records `protectedPrincipalUSD = $100,000`.
- **Step 2 (Skimming Only Accrued Interest):** At Month-End 1, the balance grows to **$100,384.95**. The option engine is restricted to skimming **only the $384.95 of newly accrued interest**, leaving 100.0% of the $100,000 floor untouched.
- **Step 3 (The High-Water-Mark Ratchet):** When a monthly Bull Call Spread finishes in-the-money and pays out **$845.53**, the net profit is swept back into the US Treasury reserve and permanently added to `protectedPrincipalUSD`. Once ratcheted into the floor, past option wins can never be lost in future months—and immediately generate additional T-Bill interest for future epochs.

Screenshot 2: PrincipalProtectedRWAVault.sol — On-Chain Floor Invariant & One-Way Ratchet

```solidity
/// @notice Skims ONLY newly accrued RWA Treasury interest; reverts if principal floor is touched
function harvestYieldToOptionEngine(bytes calldata optionOrderData) external onlyOwner returns (uint256
optionBudgetUSDC) {
    optionBudgetUSDC = harvestableOptionBudgetUSD(); // = totalRWAValueUSD() - protectedPrincipalUSD
    require(optionBudgetUSDC > 0, "NO_ACCRUED_YIELD_AVAILABLE");

    uint256 rwaSharesToSkim = rwaTreasury.convertToShares(optionBudgetUSDC);
    uint256 redeemedUSDC = rwaTreasury.redeem(rwaSharesToSkim, address(this), address(this));

    // HARD INVARIANT: Remaining RWA value must ALWAYS be >= 100% of Protected Principal Floor!
    require(totalRWAValueUSD() + 1 >= protectedPrincipalUSD, "INVARIANT_BREACH: PRINCIPAL_FLOOR_VIOLATED");
    IOptionExecutionEngine(optionEngine).executeMonthlyOptionBuy(redeemedUSDC, optionOrderData);
}

/// @notice Sweeps winning option payout back into RWA & permanently ratchets up the protected floor
function settleOptionPayout(uint256 grossPayoutUSDC) external {
    uint256 netCompoundedProfit = grossPayoutUSDC - ((grossPayoutUSDC * performanceFeeBps) / 10000);
    rwaTreasury.deposit(netCompoundedProfit, address(this));
    protectedPrincipalUSD += netCompoundedProfit; // ONE-WAY HIGH-WATER-MARK RATCHET!
}
```

2. AI Regime Filtering (TimesFM 3.0) & Dual-Track Execution

Blindly buying call options every single month underperforms plain T-Bills because of the **Volatility Risk Premium (VRP)**: during bear markets or volatility spikes, market makers overprice implied volatility. Our controller solves this via two mechanisms:

1. **Tight Bull Call Spreads (`1.01x` to `1.045x`):** Selling the upper wing offsets call skew and gives the monthly ~0.39% T-Bill interest budget **0.40x to 0.55x notional portfolio gearing** (a 2.2x–4.5x payoff on breakout months).
2. **Google TimesFM 3.0 Regime Filter:** Before deploying monthly interest, the bot runs a 20-day forward trajectory forecast using Google's TimesFM 3.0 foundation model alongside a 50-day trend and 21-day realized volatility cap. During bearish or high-vol regimes, the bot **skips buying options** and reinvests 100% of that month's T-Bill interest directly into the principal floor.

Screenshot 3: Live Execution Log — Off-Chain Brokerage Bot (SGOV + CBOE QQQ Spread)

```
$ python broker_barbell_bot.py --broker paper --symbol QQQ
[Account Base]    1,000.00 shares of SGOV (Protected Principal Floor: $100,000.00)
[Dividend Budget] $391.67 cash available from monthly SGOV T-Bill distribution

--- TimesFM 3.0 Signal & Regime Assessment ---
  Target ETF:          QQQ (Spot: $690.49)  ──►  TimesFM 20d Target: $711.95 (Expected Drift: +3.11%)
  21d Realized Vol:    22.61% (75th Pct Cap: 23.93%)  |  Signal Decision: ==> BULLISH_DEPLOY <==

--- CBOE Multi-Leg Bull Call Spread Constructed (With Real-World Slippage) ---
  Option Pair:         Buy QQQ $700.0 Call / Sell $705.0 Call (Exp: 2026-11-30)
  Theoretical Price:   $2.21/sh | Est. CBOE Bid-Ask Slippage: +$0.04/sh (1.78% drag)
  Limit Debit Order:   $2.25/sh ($225.00 for 1x 100-share contract)
  Total Dividend Used: $225.00  | Max Potential Payout: $500.00 (2.22x)
  Leftover Cash -> SGOV: +$166.67 immediately ratcheted into fractional SGOV shares!
```

On-Chain Vault vs. Off-Chain Brokerage API (Robinhood / Alpaca / IBKR)

This architecture does **not** have to run on-chain. We implemented two execution tracks in the repository:

Dimension	Track 1: On-Chain Vault (`vault_execution_bot.py`)	Track 2: Off-Chain Broker API (`broker_barbell_bot.py`)
Best Use Case	**Scaling External TVL Globally** (Smart contract collects 15% performance fee automatically)	**Compounding Personal Capital** (Zero smart-contract risk, SIPC insured, best liquidity)
Safe Treasury Base	`Sky sUSDS`, `Circle USYC`, or `Ondo USDY`	**`SGOV`** (iShares 0–3 Mo T-Bill ETF) or **`BOXX`**
Execution Venue	Hyperliquid L1 / Derive on-chain derivatives	**Robinhood (`robin_stocks`)**, **Alpaca (`MLEG`)**, or **IBKR** (CBOE Options)
Bid-Ask Slippage	**5%–15% of premium** on thinner on-chain option books	**~1.5%–2.5% of premium** ($0.02–$0.04/sh on CBOE `QQQ`/`GLD`/`IBIT`)

3. Idealized 5-Year Backtest Results (2021–2026)

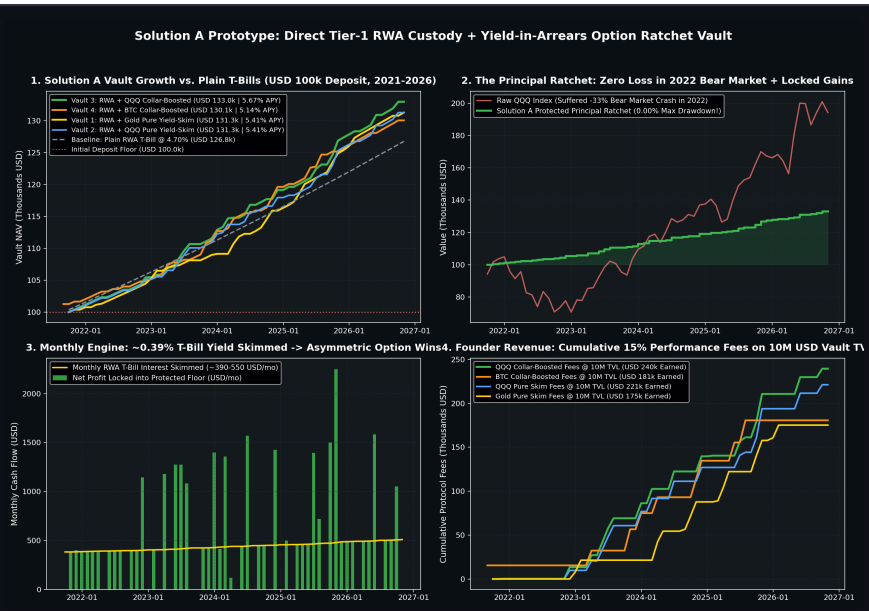


Figure 4 (4-Panel Simulation Dashboard): (1) Cumulative NAV across Gold, QQQ, and BTC Barbells vs. Plain T-Bills; (2) Step-function principal ratchet vs. 2022 QQQ crash; (3) Monthly interest skimmed vs. option payouts; (4) Cumulative 15% protocol performance fees at $10M TVL.

Simulated Strategy (2021–2026 Idealized)	Initial Deposit	Final NAV	Total Return	Idealized APY	Annual Vol	Max Principal Drawdown
Baseline: Plain US T-Bills (4.70% APY)	$100,000	$126,820	+26.8%	4.70%	0.00%	0.00%
Barbell 1: T-Bills + Gold (GLD) Pure Skim	$100,000	**$131,304**	**+31.3%**	**5.41%**	1.27%	**0.00%**
Barbell 2: T-Bills + Nasdaq (QQQ) Pure Skim	$100,000	**$131,302**	**+31.3%**	**5.41%**	1.51%	**0.00%**
Barbell 3: T-Bills + QQQ Collar-Boosted Spread	$100,000	**$132,993**	**+33.0%**	**5.67%**	1.55%	**0.00%**
Barbell 4: T-Bills + BTC Collar-Boosted Spread	$100,000	**$130,074**	**+30.1%**	**5.14%**	1.50%	**0.00%**

4. Reality Check: Idealized Backtest vs. Live Trading Slippage

> **Quantitative Disclaimer:** The 5-year simulation table above represents an **idealized theoretical backtest** using Black-Scholes pricing and continuous fractional contract sizing. Any real-world deployment faces four concrete execution frictions:

1. **Multi-Leg Bid-Ask Slippage (On-Chain vs. CBOE):** Crossing the bid-ask spread on two option legs in thin on-chain orderbooks costs **5% to 15% of the option premium**, which shaves **40 to 60 bps off annual APY**. Executing via

`broker_barbell_bot.py` on CBOE penny-pilot ETFs (`QQQ`, `SPY`, `GLD`, `IBIT`) reduces slippage to ~$0.04/sh (~1.8% of premium, or <10 bps/yr drag).

2. **The 100-Share Contract Lot-Size Constraint:** Equity options trade in 100-share lots ($225 per $5-wide `QQQ` spread). If your monthly SGOV dividend is $391.67, you can buy 1x `QQQ` spread ($225) and sweep the remaining $166.67 back into fractional SGOV shares—or trade lower-notional ETFs like **`IBIT`** (~$44/sh, where 1 spread costs ~$40–$60) for finer contract granularity.
3. **Dynamic Volatility Skew & Expiration Pin Risk:** During rapid breakouts, market makers widen OTM call implied volatility. Additionally, spreads finishing near the short strike on expiration Friday must be closed 15–30 minutes before 4:00 PM ET to avoid weekend assignment risk.
4. **Interest-Rate Cyclicality:** The simulation holds T-Bill rates at 4.70% to isolate option overlay performance. When short-term Fed rates decline toward 2.5%–3.0%, the monthly option budget shrinks proportionally.

5. Market Structure: Why Haven't Hedge Funds or Crypto VCs Built This?

A natural skeptical question is: *"If a Principal-Protected Barbell Ratchet works so cleanly, why haven't hedge funds and crypto startups flooded into it?"* It comes down to three structural realities across TradFi and Crypto:

A. Private Banks & Buffer ETFs DO Build It—Behind a Massive Fee Wall

- **Wall Street Structured Notes ($3+ Trillion Market):** Goldman Sachs, JP Morgan, and UBS sell hundreds of billions of dollars of *Principal-Protected Notes (PPNs)* using this exact math (Zero-Coupon Bond + Call Spread). However, banks embed **2.0% to 3.5% in upfront structuring fees** inside opaque CUSIPs—stealing more than half of the investor's option budget on Day 1.
- **Defined-Outcome / Buffer ETFs:** Issuers like Calamos (`CPSM`, `CPNQ`, `CPBI`) and Innovator have raised tens of billions in 100%-downside-protected ETFs since 2024, but they are locked into static 1-year expiries without monthly ratchets or AI regime filters.

B. Why Hedge Funds Can't Touch It: The "2-and-20" Fee Math Problem

- A Manhattan hedge fund paying 10 PhDs, prime brokers, and office rent requires a **2% management fee + 20% performance fee**.
- If a low-volatility Barbell generates **5.67% gross APY** (+97 bps of pure alpha over 4.70% T-Bills), deducting a 2% management fee + 20% performance fee leaves the client with **~3.3% net APY—worse than a plain T-Bill!**
- Institutional LPs refuse to pay hedge fund fees on 99.6% T-Bill custody. Thus, a 5.5%–6.0% APY zero-loss Barbell is economically unviable for a bloated human hedge fund—**it only works when automated by zero-overhead software or run on your own capital with 0% fees.**

C. Why Crypto Startups Missed It: The 2009–2022 Zero-Rate Era & "Degen Marketing"

- **Zero Rates During DeFi Summer:** From 2009 to 2022, US T-Bills yielded **0.05% to 0.25%**. You cannot build a T-Bill Yield-Skim Option Vault when $100,000 generates $100/year in interest. That forced early DeFi Option Vaults (Ribbon, Friktion) to sell naked short options instead—which blew up in the 2022 crash. This strategy only became viable on-chain once T-Bills hit 4.5%+ and Tier-1 tokenized Treasuries (`sUSDS`, `BUIDL`, `USYC`) scaled in 2024–2026.
- **Crypto Marketing Culture:** Venture-backed DeFi protocols feel pressured to advertise "25% APY" to attract mercenary retail capital, leading them into fragile leverage loops (`xUSD` / `deUSD`) rather than sustainable 5.5%–6.5% principal-protected ratchets.

> **Bottom Line for Practitioners:** This strategy sits in a structural sweet spot for an individual quant or lean software builder. By automating **`SGOV` + CBOE Bull Call Spreads via Robinhood/Alpaca API** (for personal capital) or an **ERC-4626 RWA Vault** (for zero-overhead protocol scaling), you capture the exact payoff profile of a private-bank structured note while keeping 100% of the structuring margin in your own pocket.